Healthcare Data Analytics — Business Problem & Deliverables

Problem Statement

Healthcare admission records contain useful information about patient demographics, medical conditions, admission types, billing, insurance, and hospital stay duration. However, duplicate records, inconsistent text formats, and unusual billing values can reduce the reliability of analysis. This project will clean and validate the available dataset, then analyze admission patterns, recorded billing amounts, and length of stay. The goal is to turn the data into clear, evidence-based insights that may support healthcare operations reporting. The dataset does not include employee data or support claims about treatment effectiveness or readmission rates.

Deliverables

1. Data Cleaning & Validation (Python — Pandas and NumPy)

Inspect data types, missing values, duplicate rows, inconsistent text, date fields, and unusual billing values. Document each cleaning decision and preserve the raw file.

2. Data Transformation

Create useful analysis fields where supported, including patient age groups and length of stay calculated from admission and discharge dates. Validate derived values before use.

3. Exploratory Data Analysis (EDA)

Explore patient demographics, medical-condition counts, admission types, billing distributions, insurance-provider comparisons, and length-of-stay patterns using clear charts.

4. SQL Analysis (MySQL)

Load the cleaned data into MySQL and write queries using filtering, aggregation, grouping, CASE statements, date calculations, and ranking to answer the project questions.

5. Interactive Power BI Dashboard

Build a concise dashboard showing key admission counts, billing summaries, length-of-stay patterns, and relevant breakdowns by medical condition, admission type, and insurance provider. Validate dashboard totals against Python and SQL results.

Expected Outcome

A documented and validated dataset, reproducible Python analysis, SQL query file, and interactive Power BI dashboard. Findings will describe patterns observed in this dataset only; they will not be presented as causal or clinically validated conclusions.

Dataset Source

Kaggle: https://www.kaggle.com/datasets/prasad22/healthcare-dataset